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MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

1/ TYPE 8939

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Coaxial, tunable frequency 8,500 to 9,600 MHz, pulse type, integral magnet, 250-kw rated peak power output

ABSOLUTE RATINGS:

Parameter:	Ef	If(surge)	tk	tpc	rrv	Du	epy	ib	VSWR
Unit:	V	a	sec	us	kv/us	---	kv	a	---
Maximum:	15	12	---	0.35	200	0.001	28	30	1.5:1
Minimum:	---	---	150	0.18	90	---	---	15	---
	(Note 1)				(Note 2)				

ABSOLUTE RATINGS:

			Temperature		Tuner		Pressurization	
Parameter:	Pi	pi	Body	Input bushing	stop torque	Alt	Input	Output
Unit:	W	kw	°C	°C	in.-oz	ft	psia	psia
Maximum:	600	800	130	165	48	10,000	45	45
Minimum:	---	---	-55	-55	---	---	15	15
			(Note 3)	(Note 3)	(Note 4)		(Note 5)	

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 8, 28, and note h (fig 1)
Mounting position:	Any		
Mounting support:	See figure 1 (flange)	Cooling:	Forced air (see note 9)
Input coupling:	See notes 6, 7 and See figure 1	Magnet isolation:	See note 10
Output coupling:	See notes 6, 7 and See figure 1	Weight:	17 pounds (approx)

TEST CONDITION (1): See notes 6 and 7

Parameter:	Ef	tk	tpc	rrv	Du	Ib	VSWR	F
Unit:	V	sec	us	kv/us	---	mAdc	---	MHz
Maximum:	---	---	0.33	---	---	---	---	---
	13.75	---	---	---	0.0004	11	1.1:1	F1, F2, F3
Minimum:	---	150	0.27	180	---	---	---	---
	(Note 1)							

FREQUENCY		
F	MHz	±
1	8,500	5
2	9,000	5
3	9,600	5

TEST CONDITION (2): See notes 6 and 7

Parameter:	Ef	tk	tpc	rrv	Du	Ib	VSWR	F
Unit:	V	sec	us	kv/us	---	mAdc	---	MHz
Maximum:	---	---	0.33	---	---	---	---	---
	13.75	---	---	---	0.0006	16.5	1.1:1	F1, F2, F3
Minimum:	---	150	0.27	180	---	---	---	---
	(Note 1)			(Note 2)				

GENERAL:

Qualification - Required

1/ Replaces Varian type SFD-382.

Ⓑ 2/ See note 28

Ⓑ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4223	Torque	-	-	T (body) = -55°C nonoperating to +130°C operating				
	Starting	-	2		---	---	12	in.-oz
	Operating	-	2		---	---	10	in.-oz
4223	Mechanical tuning fatigue	12	-	No voltages applied	---	50,000	---	Cycles
1031	Low-frequency vibration	13,14	-	No voltages applied	---	---	---	---
1031	High-frequency vibration	13,14	-	No voltages applied	---	---	---	---
4027	Temperature coefficient	3,15	1	F = F2; T = 70° to 130°C	$\Delta F/\Delta T$	---	0.25	MHz/°C
1042	Shock, specified pulse	14,16	-	Condition J	---	---	---	---
1006	Salt spray (corrosion)	17,18,19	-	Method B	---	---	---	---
4551	Life test	26	-		t	2000	---	hrs
---	life test end points:							
4250	Power output	-	1		Po	80	140	W
4308	RF bandwidth	-	1		BW	---	2.5/tpc	MHz
4308	Minor lobe ratio	-	1		---	6	---	dB
4315	Stability	-	1	t = 180 seconds	MP	---	0.5	%
4223	Resettability	27	1		ΔF	---	15	MHz
	Quality conformance inspection, part 1	20	-					
4003	Pressurizing	-	-	45 psia minimum, input and output	---	---	---	---
4289	Heater current (nonoperating)	-	-	Ef = 13.75 V	If	2.9	3.3	A
4303	Warmup time	1	1,2		---	---	---	---
4306	Pulse voltage	-	-		epy	23	27	kv
4250	Power output	-	1		Po	100	140	W
4308	RF bandwidth	-	1		BW	---	2.0/tpc	MHz
4308	Minor lobe ratio	-	1		---	10	---	dB
4315	Stability	-	1	t = 180 seconds	MP	---	0.25	%
4315	Starting stability	21	1	F = F3	MP	---	0.25	%
4223	Operating torque	-	-	F1 thru F3; no voltages applied	---	---	10	in.-oz
4223	Mechanical tuning range	22	1	Upper limit	F	9,605	---	MHz
				Lower limit	F	---	8,495	MHz

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REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
Quality conformance inspection, part 2							
Forced cooling	3,9,23	2		ΔT (body)	---	90	°C
Frequency pushing figure	24	2		ΔF/Δib	---	125	kHz/a
Frequency pulling figure	-	2		ΔF	---	6.0	MHz
Resettability	25	2	F1 +50 ±5 MHz; F2 ±5 MHz; F3 -50 ±5 MHz	ΔF	---	3	MHz
Direct-interelectrode capacitance	-	-	Cathode terminal to mounting plate	C	9	14	pF
Shelf life	11	---	90 days	---	---	---	---
Quality conformance inspection, part 3							
Life test	26	-	Group D	t	500	---	hrs
life test end points:							
Power output	-	1		Po	80	140	W
RF bandwidth	-	1		BW	---	2.5/tpc	MHz
Minor lobe ratio	-	1		---	6	---	dB
Stability	-	1	t = 180 seconds	MP	---	0.5	%
Resettability	27	1		ΔF	---	15	MHz

Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This shall be done by applying 13.75 volts ±5 percent for 150 seconds minimum. Upon the application of anode voltage, the heater voltage shall be reduced according to the following schedule:

	Du	Ib (mAdc)	Ef (V) -5%
Standby:	---	---	13.75
Test condition 1:	0.0004	11.0	9.0
Test condition 2:	0.0006	16.5	6.0

(1) For Pi between 0 and 375 W, Ef = 13.75 (1-Pi/800) -5%
(2) For Pi between 375 W and 600 W, Ef = 19 (1-Pi/600) -5%

The characteristics of the applied pulse shall be those which result in proper starting and oscillation. The rate of rise of voltage (rrv) shall be measured in accordance with method 4304, except that the steepest tangent to the leading edge of the voltage pulse shall be measured above the 70 percent amplitude point.

The temperature is to be measured at the point indicated on figure 1 (see notes e and f).

The tuning mechanism shall be capable of withstanding a static torque of 48 in.-oz at the end of its travel.

The tube under test (TUT) shall be capable of normal operation without electrical breakdown with the input bushing in air at normal atmospheric conditions.

The minimum air pressure to assure prevention of electrical breakdown in the output coupling shall be 15 psig for voltage standing wave ratios up to 1.5:1 with phase shift variable over 360 electrical degrees. The tube shall be coupled directly to a UG-52/U choke flange modified so that mounting holes provide clearance for No. 8 bolts.

The modulator shall be such that energy per pulse delivered to the tube, if arcing occurs, shall not greatly exceed the normal energy per pulse. The tube heater shall be protected against arcing by use of a connector that places a minimum of 4,000 pF across the heater directly at the input terminals.

In addition to regular markings, the tuner dial settings for the following frequencies shall be marked on the tube during final testing by the manufacturer. The accuracy of these settings shall be ±10 MHz at the start of life under conditions of test condition 2 with the body temperature approximately 80°C measured at the point specified on figure 1 when tuning is performed in the order of decreasing the frequency.

Frequency (MHz)	Dial setting	Frequency (MHz)	Dial setting.
8,500	---	9,150	---
8,650	---	9,300	---
8,800	---	9,450	---
9,000	---	9,600	---

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NOTES: -Continued

9. With a total airflow of approximately 40 cfm at approximately 760 mmHg, divided equally and directed through the cooling fins toward the body of the tube from two 3/4-inch (19.05 mm) ducts placed 1/2 to 3/4 inch (12.7 to 19.05 mm) from the cooling fins, the rise above ambient temperature specified herein shall not be exceeded.

10. In handling and mounting the tube, care shall be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 4 inches (101.6 mm) of the tube.

(E) 11. This test shall be performed on four tubes per month when tube is in continuous production, but shipments of the month's production shall not be held pending completion of the test. So long as three of the four tubes for each of the first three months of a production run pass the test, and 75 percent of the cumulative quantity of tubes tested pass the test, tubes shall be considered to conform to this specification. If either of the conditions are not met, shipments shall be halted until three of four tubes of current production conform to test.

12. The tuning shaft shall be continuously driven at a speed of 1,400 ±50 rpm. At the completion of the test the backlash shall not exceed 5 MHz. During the test, the tuning shaft may not be lubricated. A cycle consists of two complete excursions each in opposite directions through the tuning range of the tube.

13. The tube shall be vibrated in three axes, perpendicular to reference planes A, B, and C, figure 1, for a period of 1 minute in each plane.

14. At the completion of this test, the tube shall meet the power output, voltage, and tuner torque requirements of test condition 1 of quality conformance inspection, part 1.

15. Temperature measurements shall be made only after thermal equilibrium has been reached. The frequency shall be measured at the extremes of any 30°C temperature range.

16. The tube shall be subjected to five impact shocks along each of three axes perpendicular to reference planes A, B, and C (figure 1). Each shock pulse shall be half-sine shape, and have a duration of 11 ±1 milliseconds. The mounting plate of the tube shall be bolted with brass bolts to the supporting fixture, and the shock pulse shall be measured on this fixture near the mounting plate.

17. The salt solution concentration shall be 20 percent. The tuning mechanism may be completely cycled (cw and ccw) once prior to measuring static torque. The tube shall meet the requirements of quality conformance inspection, part 1, after salt spray.

18. The waveguide output flange and the high-voltage input bushing shall be enclosed for the salt spray test.

19. At the conclusion of this test, the tuner starting torque shall not exceed 14 in.-oz.

20. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

21. After the holding period, the anode voltage shall be applied 150 seconds minimum after the application of the standby heater voltage. The missing pulse count test interval of 3 minutes shall start immediately after the application of anode voltage.

22. The frequency range F1 to F3 shall be traversed by a tuning shaft rotation of 160 ±4 turns. A mechanical over-run of 3 turns minimum shall be provided at each end of the tuning range before contacting the mechanical stops.

23. The frequency shall be the frequency of minimum power output between F1 and F3.

24. The pushing factor shall be measured in a 4 ampere step, 24 to 28a, and shall not exceed the limits specified herein. The peak current through the tube shall alternately be the limits as specified under this test condition. These tests shall be run to exclude the effects of thermal drift and frequency instability not due to pushing.

25. The frequency obtained by tuning to a given angular position in one direction shall be reproducible within the specified limits when returning to that same position from the opposite direction after thermal equilibrium.

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NOTES: -Continued

26. The life test shall be conducted while the phase of a 1.5:1 minimum VSWR is cycled through 180 electrical degrees minimum, approximately every 15 minutes. Simultaneously the tube frequency shall be cycled, or changed in 150 MHz increments, each 50 hours to insure approximately equal operating time over the entire frequency range from F1 to F3. After the required standby condition, the following test cycle shall apply:

Condition	Ib (mAdc)	Ef (V)	Duration (minutes)
Standby	0	13.75	10
Test condition 1	11	9.0	110

Note: The accumulated time shall include both high voltage and standby time.

27. At each of the dial settings and under the operating conditions specified in note 8, the measured frequency shall not differ from the stated frequency by more than the amount specified when tuning is performed in the order of increasing the frequency.

Ⓡ 28. Each magnetron shall be tested at an RF output of 300 kw at 9.0 GHz with a pulse width of 0.3 microseconds, at duty cycles of 0.0003 and 0.0004. The magnetron shall be marked as follows:

Duty cycle	Ib (mAdc)
0.0003	-------
0.0004	-------

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - AR
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3159)

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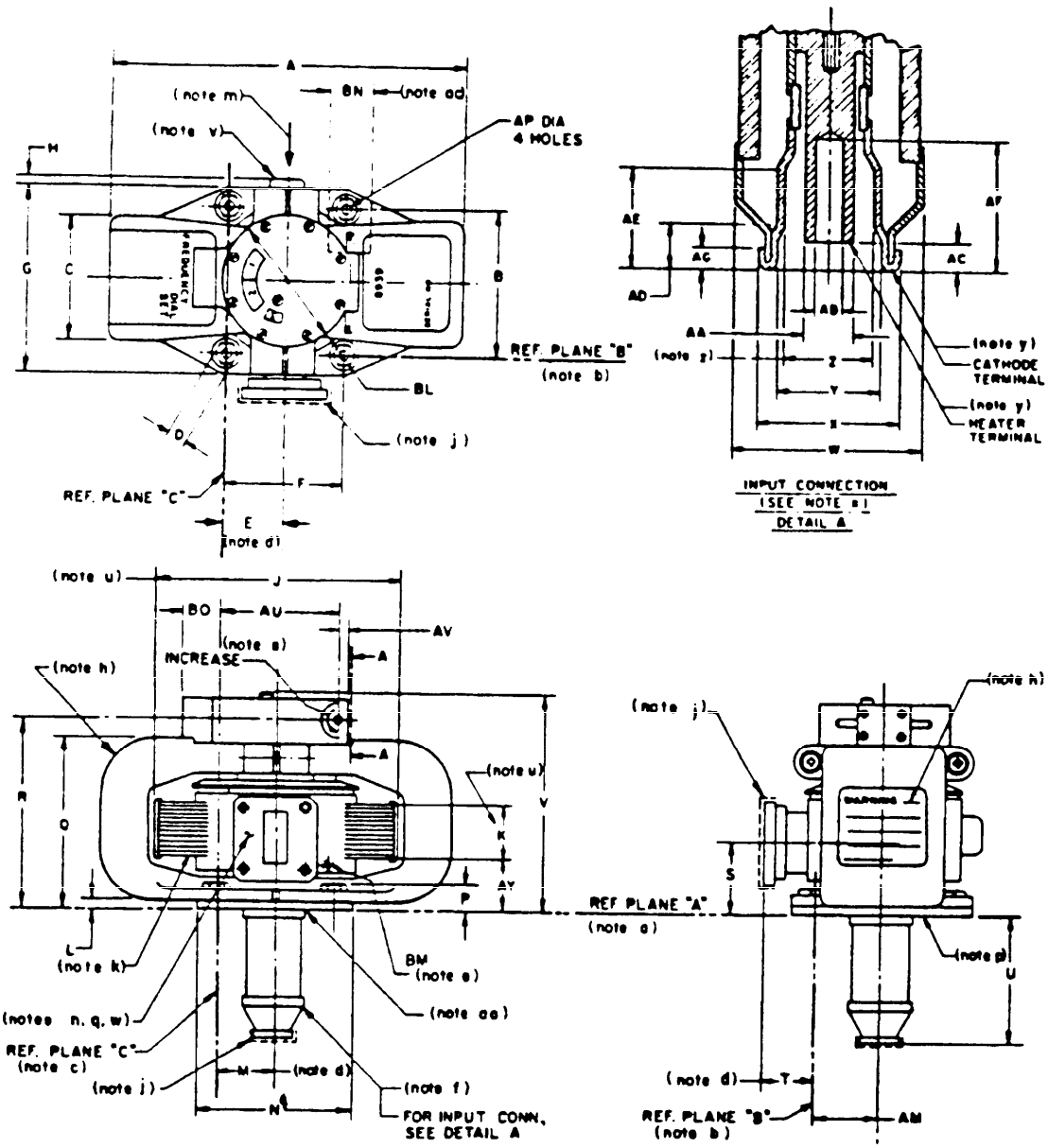
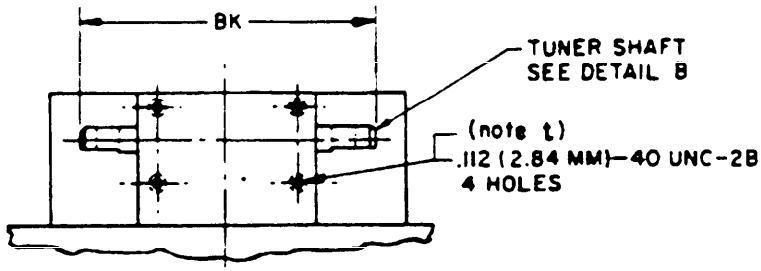
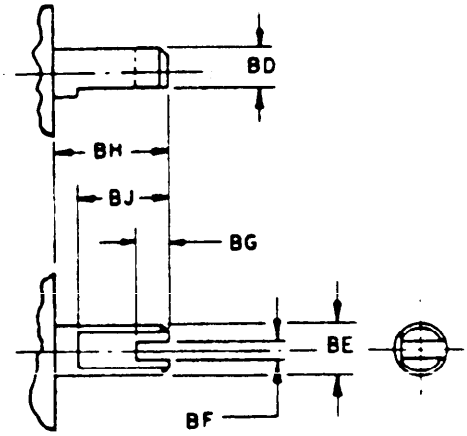


FIGURE 1. Outline drawing of electron tube type 8939.

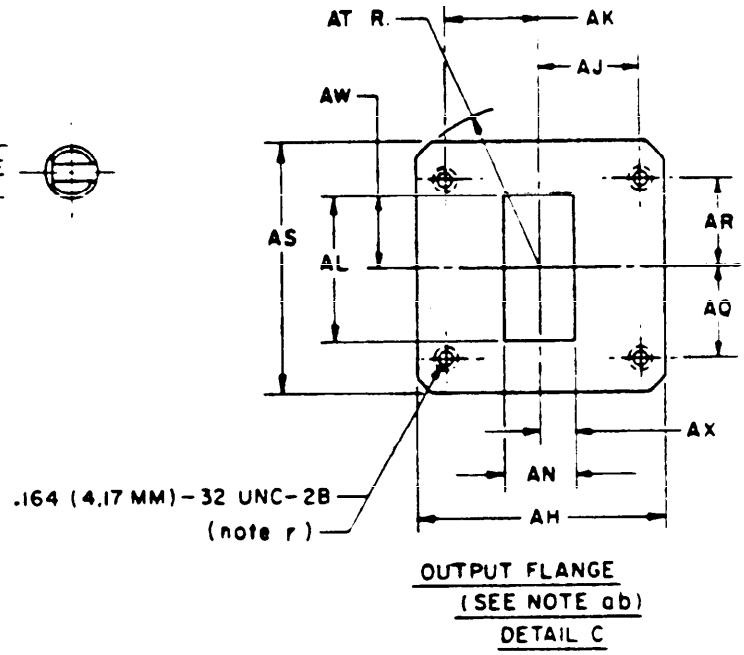
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VIEW A-A



TUNER SHAFT
DETAIL B



OUTPUT FLANGE
(SEE NOTE ab)
DETAIL C

FIGURE 1. Outline drawing of electron tube type 8939 - Continued.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Qualication inspection		
A	---- ----	9.400 (238.76)
C	---- ----	2.780 (70.61)
D	.470 (11.94)	.530 (13.46)
L	.031 (.79)	---- ----
M	1.204 (30.58)	1.296 (32.92)
Q	---- ---	3.900 (99.06)
V	---- ----	5.000 (127.00)
Y	.603 (15.32)	.617 (15.67)
AD	---- ----	.156 (3.96)
AE	.562 (14.27)	---- ----
AG	.115 (2.92)	.135 (3.43)
AH	1.820 (46.23)	1.840 (46.74)
AL	1.119 (28.42)	1.125 (28.58)
AM	1.454 (36.93)	1.546 (39.27)
AN	.494 (12.55)	.500 (12.70)
AS	1.820 (46.23)	1.840 (46.74)
AV	.391 (9.93)	.421 (10.69)
AW	.5595(14.211)	.5625 (14.288)
AX	.247 (6.27)	.250 (6.35)
BF	.038 (.97)	.042 (1.07)
BG	.120 (3.05)	.130 (3.30)
BJ	.333 (8.46)	.353 (8.97)
Quality conformance inspection, part 1		
B	2.990 (75.95)	3.010 (76.45)
E	1.204 (30.58)	1.296 (32.92)
F	2.490 (63 25)	2.510 (63.75)
R	4.310 (109.47)	4.434 (112.62)
S	1.829 (46.46)	1.921 (48.79)
T	1.003 (25.48)	1.067 (27.10)
U	2.381 (60.48)	2.477 (62.92)
X	.825 (20.96)	.838 (21.29)
AC	.125 (3.18)	.187 (4.75)
AF	.750 (19.05)	---- ----
AU	2.440 (61.98)	2.556 (64.92)
BH	.428 (10.87)	.448 (11.38)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	MINIMUM	MAXIMUM
Quality conformance inspection, part 2		
W	1.062 (26.97)	1.125 (28.58)
Z	.532 (13.51)	.545 (13.84)
AA	.234 (5.94)	.266 (6.76)
AB	.164 (4.17)	.174 (4.42)
AJ	.735 (18.67)	.739 (18.77)
AK	.735 (18.67)	.739 (18.77)
AP	.276 (7.01)	.286 (7.26)
AQ	.674 (17.12)	.678 (17.22)
AR	.674 (17.12)	.678 (17.22)
BD	.153 (3.89)	.159 (4.04)
BE	.1855 (4.712)	.1885 (4.788)
BL	2.625 (66.68)	2.750 (69.85)
BN	.505 (12.83)	.525 (13.34)
BK	2.240 (56.90)	2.260 (57.40)
	Nominal dimensions	
G	3.750 (95.25)	
H	.297 (7.54)	
J	5.400 (137.16)	
K	1.200 (30.48)	
N	3.390 (86.11)	
P	.581 (14.76)	
AT	1.150 (29.21)	
AY	1.400 (35.56)	
BM	4-40 NC X .250 (6.35) deep min.	
BO	.800 (20.32)	

FIGURE 1. Outline drawing of electron tube type 8939 - Continued.

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NOTES:

a. Reference plane "A" is defined as a plane passing along the face of the mounting plate.
b. Reference plane "B" is defined as a plane perpendicular to plane "A" passing through the axis of the holes, as shown at reference plane "A".
c. Reference plane "C" is defined as a plane mutually perpendicular to planes "A" and "B" passing through the axis of the holes, as shown at reference plane "A".
d. Includes angular as well as lateral deviations. Dimension E is to centerline (℄) of waveguide and dimension M to ℄ of heater terminal.
e. Anode temperature to be measured at this point.
f. Input bushing temperature to be measured at this point.
g. For vibration of shock testing, the planes of testing shall be reference planes "A" - "B" - "C".
h. Warning: Maintain minimum clearance of 4 inches between magnet and magnetic materials (magnets, steel tools, plates, etc). Use no magnetic inspection tools.
j. Protective closure. To be removed before tube is used, and attached when the tube is not in use.
k. The inner laminations of the cooling fins are not painted. However, there may be an over-spray protective paint.
m. Indicates direction of body cooling airflow.
n. Mates with modified choke flange, UG-52 B/U (MIL-F-3922/59) (clearance instead of threaded holes).
p. With the tube mounted on the test fixture and with the specified air pressure applied so as to surround the entire input terminal beyond the magnetron mounting plate, the entire magnetron and fixture are to be submerged in water. No bubbles are allowed in a 1-minute interval. An equivalent test method to insure adequate air pressure containment is acceptable.
q. With the tube output flange clamped against a modified choke flange (UG-52 B/U) and using a gasket per MS-90064-14, and with the specified air pressure applied to the interior of the waveguide submerge entire tube and fixture in water. No bubbles are allowed in a 1-minute interval. An equivalent test method to insure adequate air pressure containment is acceptable.
r. A plane passing through the axis of two threaded holes perpendicular to the face of the output flange shall be parallel to planes "A" and "C" within .030 (.76 mm).
s. The frequency increases as the tuner input shaft is driven in the direction indicated by the arrow.
t. The absence or presence of these holes does not impair serviceability of the tube.
u. This dimension refers to radiator fin size only.
v. Permanently attached protective closure for the exhaust tubulation of the tube will not restrict airflow nor impair serviceability of the tube.
w. The face of the waveguide flange shall be flat within .010 (.25 mm) total indicator reading. The surface of this flange shall be of such a quality that a hermetic type seal can be effected.
x. Input connection mates with Jettron Products Inc., Hanover, N.J. Connectors Catalog No. 90-006 and 90-030, or equivalent.
y. Heater terminal and cathode terminal shall be concentric within .010 (.25 mm).
z. This diameter applies for length "AE" minus "AD".
aa. This identifies a north seeking pole. This may be tested by observing which compass needle points to the geographic south. The same compass needle will point toward the tube's north seeking pole when it is brought near the input stem of the tube. Care should be taken to insure that the tube field does not reverse the polarity of the compass.
ab. The face of the waveguide flange is 70-30 cupronickel. (Copper Development Assn. Specification No. CDA-715, or equivalent). This face has a satin type (smooth, dull) finish of approximately 63 microinches. Galvanic couples for this material are listed in MIL-E-16400 (NAVY).
ac. This diameter identifies clearance area for mating shaft.

FIGURE 1. Outline drawing of electron tube type 8939. - Continued

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